

STOCHASTIC GRADIENT LANGEVIN DYNAMICS

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ABSTRACT

Standard Stochastic Gradient Langevin Dynamics (SGLD) relies on Gaussian noise, which often suffers from metastability and exponentially slow escape times from deep local minima. While recent theories establish dimension independent generalization bounds using temperature scaling, they assume worst case Gaussian structures, yielding conservative limits. Conversely, heavy-tailed α -stable Lévy processes, driven by discontinuous jumps, are known to drastically accelerate convergence in non convex landscapes. In this report, we survey the mathematical foundations of fractional Langevin dynamics and temperature dependent generalization guarantees. Based on this literature review, we outline a theoretical proposal for deriving tighter, tail aware generalization bounds. We hypothesize that the rapid exit times inherent to Lévy jumps fundamentally reduce the algorithmic reliance on high temperatures.

1 INTRODUCTION

While Stochastic Gradient Descent (SGD) is the standard for optimizing deep neural networks, it generally seeks Maximum A Posteriori (MAP) estimates, often getting trapped in local minima or saddle points in high-dimensional, non-convex landscapes. Stochastic Gradient Langevin Dynamics (SGLD) is motivated by the need to transition from single-point optimization to Bayesian posterior sampling, allowing for uncertainty quantification and better generalization. As proposed by Welling & Teh (2011), by injecting Gaussian noise scaled by the step size into the parameter updates, SGLD enables the trajectory to explore the parameter space rather than collapsing to a single mode. In non-convex settings, as analyzed by Raginsky et al. (2017), this injected noise facilitates the escape from local minima (metastability) and drives the distribution of the iterates toward the stationary Gibbs distribution. However, the convergence rate is not uniform; it is intrinsically linked to the geometric properties of the objective function $U(X)$. Specifically, the convergence speed is governed by the spectral gap (or the constant in the log-Sobolev inequality) of the underlying continuous-time diffusion process. When the energy landscape is rugged with high potential barriers, the spectral gap becomes exponentially small with respect to the inverse temperature parameter β , leading to slow mixing times as the process struggles to traverse between different modes of the distribution.

As previously mentioned, in the context of non-convex optimization, standard Stochastic Gradient Langevin Dynamics (SGLD) relies on Gaussian noise, which corresponds to driving the system with standard Brownian motion. In a statistical physics context, X_t often represents

the position of a particle (at time t) that is under the influence of a random force. In this case, the Langevin equation is motivated by the hypothesis that this random force is the sum of many i.i.d. random pulses, whose variance is assumed to be finite (Yanovsky et al. (2000)). By the Central Limit Theorem (CLT), the sum of these pulses converges to a Gaussian random variable, which justifies the choice of the Brownian motion.

However, recent studies in deep learning demonstrate that the stochastic gradient noise often exhibits infinite variance, violating this finite-variance assumption (Şimşekli et al. (2019)). Consequently, if we relax the finite variance assumption, the usual CLT would not hold. Instead, the sum of these random pulses converges to a class of heavy-tailed distributions known as α -stable (or Lévy-stable) distributions.

A limitation of Brownian motion is its almost surely continuous paths. As opposed to the Brownian motion, the Lévy motion can contain discontinuities that are often referred to as jumps. Instead of continuously climbing steep energy barriers, the optimization trajectory can leverage these heavy-tailed jumps to leap directly out of deep valleys. Recent studies have established that utilizing such heavy-tailed dynamics yields faster convergence rates for non-convex functions (Şimşekli (2017)).

Recent theoretical frameworks have shifted the focus from the asymptotic sampling properties of Langevin dynamics to robust, non-asymptotic generalization guarantees. A significant development in this direction is the work by (Harel et al., 2025), which establishes that the generalization gap can be bounded at any intermediate step of the training process. Unlike classical analyses that rely on the mixing time of the Markov chain to reach a stationary pos-

terior, a process that is often exponentially slow in high-dimensional settings, this approach demonstrates that the mere existence of a Gibbs-type stationary distribution is sufficient to regulate model complexity. By leveraging a generalized second law of thermodynamics, it is shown that the divergence of the marginal distribution from its initialization remains monotonic and controlled, resulting in generalization bounds that are independent of parameter dimensionality, training duration, or gradient norms.

Building upon these time-invariant guarantees, we propose a transition from standard Gaussian-driven dynamics to heavy-tailed stochastic processes to address the fundamental challenge of metastability. While Gaussian noise is effective for exploring smooth landscapes, it often fails to efficiently overcome the barriers of deep local minima in the highly non-convex surfaces typical of large-scale models. We hypothesize that moving toward a Fractional Langevin framework, utilizing Lévy-driven noise, allows for a more noise-aware generalization bound. By exploiting the power-law characteristics of α -stable distributions, this approach facilitates faster exit times from suboptimal minima, potentially offering a tighter and more technically descriptive bound for generalization in modern high dimensional architectures compared to traditional Brownian motion.

2 BACKGROUND

2.1 STANDARD SGLD AND METASTABILITY

The theoretical framework of Stochastic Gradient Langevin Dynamics (SGLD) rests on the discretization of the continuous Langevin diffusion process, a stochastic differential equation (SDE) that models Brownian motion under the influence of a potential energy field. Let $U : \mathbb{R}^d \rightarrow \mathbb{R}$ be the non-convex objective function, representing the negative log-posterior in a Bayesian setting. The continuous-time dynamics of the parameter vector X_t are governed by the SDE:

$$dX_t = -\nabla U(X_t)dt + \sqrt{2\beta^{-1}}dB_t, \quad (1)$$

where B_t is a standard d -dimensional Brownian motion and β is the inverse temperature parameter. Under standard regularity conditions, the stationary distribution of this process is the Gibbs measure $\pi(dX) \propto e^{-\beta U(X)}$. To apply this to large-scale datasets, Welling & Teh (2011) proposed the SGLD algorithm, which approximates the continuous path using a sequence of discrete updates with stochastic gradients. At iteration k , utilizing a step size η_k and a stochastic gradient estimate $g_k(X_k)$ (where $\mathbb{E}[g_k(X_k)] = \nabla U(X_k)$), the update rule is defined as:

$$X_{k+1} = X_k - \eta_k g_k(X_k) + \sqrt{2\eta_k \beta^{-1}} \xi_k, \quad \xi_k \sim \mathcal{N}(0, I_d). \quad (2)$$

The convergence properties of this discrete algorithm in non-convex landscapes have been rigorously analyzed by Raginsky et al. (2017) using the Wasserstein-2 distance,

$W_2(\mu_k, \pi)$, to measure the deviation of the algorithm's finite-time distribution μ_k from the target posterior π . The analysis establishes that the convergence rate is fundamentally governed by the spectral gap λ_* of the underlying diffusion operator. Specifically, the upper bound on the Wasserstein distance at step k generally follows a structure where the error is dominated by two terms:

$$W_2(\mu_k, \pi) \lesssim \underbrace{C_1 e^{-\lambda_* k \eta}}_{\text{Geometric Decay}} + \underbrace{C_2 \sqrt{\eta}}_{\text{Discretization Noise}}. \quad (3)$$

This relation illustrates a trade off between the bias introduced by time discretization (controlled by the step size η) and the mixing time of the process (controlled by the total simulation time $T = k\eta$). To achieve a target accuracy ε in the W_2 metric, the step size must scale polynomially with ε , while the number of iterations k must scale inversely with the spectral gap, approximately $\mathcal{O}(d \cdot \text{poly}(1/\varepsilon) \cdot \lambda_*^{-1})$. Critically, for non-convex functions with multiple modes, the spectral gap λ_* decays exponentially with the height of the energy barriers and the inverse temperature β (i.e., $\lambda_* \propto e^{-\beta H}$), implying that while the dependence on dimension d is polynomial, the time required to traverse the energy landscape and reach the stationary distribution can be exponential in the problem parameters if the landscape is rugged.

2.2 HEAVY-TAILED DYNAMICS AND LÉVY JUMPS

Stable distributions are heavy-tailed distributions. They are the limiting distributions in the generalized central limit theorem: the properly scaled sum of i.i.d. random variables, which do not necessarily have finite variance, will converge to an α -stable random variable Samorodnitsky & Taqqu (1994). In this study, we are interested in the centered symmetric α -stable ($S\alpha S$) distribution, which is a special case of α -stable distributions.

The $S\alpha S$ distribution can be seen as a heavy-tailed generalization of the centered Gaussian distribution. The probability density function (pdf) of an $S\alpha S$ distribution cannot be written in closed-form except for certain special cases; however, the characteristic function of the distribution can be written as follows:

$$X \sim S\alpha S(\sigma) \iff \mathbb{E}[\exp(i\omega X)] = \exp(-|\sigma\omega|^\alpha).$$

Here, $\alpha \in (0, 2]$ is called the characteristic exponent and determines the tail thickness of the distribution: as α gets smaller, $S\alpha S$ becomes heavier-tailed. The parameter $\sigma \in \mathbb{R}_+$ is called the scale parameter and measures the spread of X around 0.

As an important special case of $S\alpha S$, we obtain the Gaussian distribution $S\alpha S(\sigma) = \mathcal{N}(0, \sigma^2)$ for $\alpha = 2$. In Figure 1, we illustrate the (approximately computed) pdf of the symmetric α -stable distribution for different values of α . As can be clearly observed from the figure, the tails of the distribution vanish quickly when $\alpha = 2$ (i.e. Gaussian), whereas the tails get thicker as we decrease α . Consequently, the size of these jumps becomes larger as α gets smaller, since $S\alpha S$ becomes heavier tailed.

SDEs driven by symmetric stable Lévy processes: In this study, we are interested in SDEs driven by symmetric α -stable Lévy processes, which are defined as follows:

$$dX_t = b(X_{t-}, \alpha)dt + dL_t^\alpha,$$

where L_t^α is defined by the following properties:

- (i) $L_0^\alpha = 0$ almost surely.
- (ii) For $t_0 < t_1 < \dots < t_N$, the increments $(L_{t_n}^\alpha - L_{t_{n-1}}^\alpha)$ are independent ($n = 1, \dots, N$).
- (iii) The difference $(L_t^\alpha - L_s^\alpha)$ and L_{t-s}^α have the same distribution: $S\alpha S((t-s)^{1/\alpha})$ for $s < t$.
- (iv) L_t^α has stochastically continuous sample paths (i.e. continuous in probability): for all $\delta > 0$ and $s \geq 0$, $\mathbb{P}(|L_t^\alpha - L_s^\alpha| > \delta) \rightarrow 0$ as $t \rightarrow s$.

These sample paths are continuous from the right and they have left limits at every time. Therefore, the term X_{t-} in the SDE explicitly denotes this left limit.

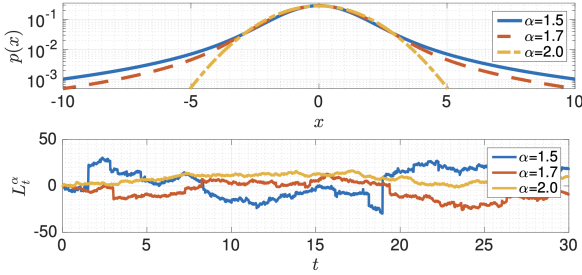


Figure 1: The pdf of the symmetric α -stable distribution for different values of α . The tails become heavier as α decreases.

Our primary goal is to find a drift function b such that the Markov process $(X_t)_{t \geq 0}$, which is a càdlàg solution to the SDE, has the target distribution π as an invariant distribution Şimşekli (2017). This is formally established in the following theorem:

Theorem 1. Consider the SDE $dX_t = b(X_{t-}, \alpha)dt + dL_t^\alpha$, where the drift b is defined as:

$$b(x, \alpha) \triangleq \frac{\mathcal{D}^{\alpha-2} f_\pi(x)}{\phi(x)}. \quad (4)$$

Here, $f_\pi(x) \triangleq -\phi(x)\partial_x U(x)$. Then, π is an invariant measure of the Markov process $(X_t)_{t \geq 0}$ that is a càdlàg solution of the given SDE. Furthermore, if b is Lipschitz continuous, then π is the unique invariant measure of the process.

Theorem 1 suggests that continuous sample paths from this process could serve as samples drawn directly from

π ; however, simulating the continuous time SDE exactly is generally not possible. Therefore, we focus on simulating the SDE by considering its Euler-Maruyama discretization, given as follows:

$$\bar{X}_{n+1} = \bar{X}_n + \eta_{n+1}b(\bar{X}_n, \alpha) + \eta_{n+1}^{1/\alpha}\Delta L_{n+1}^\alpha, \quad (5)$$

where $n = 1, \dots, N$ denotes the time-steps, $(\eta_n)_n$ is a sequence of step-sizes, and $(\Delta L_n^\alpha)_n$ is a sequence of i.i.d. standard symmetric α -stable random variables, i.e., $\Delta L_n^\alpha \sim S\alpha S(1)$.

To ensure that the estimator obtained from this discretized scheme properly converges to the true expectation under the target density, it has been established that certain conditions must be satisfied Şimşekli (2017). The primary requirement is a standard assumption on the step-sizes: $\lim_{n \rightarrow \infty} \eta_n = 0$ and $\lim_{N \rightarrow \infty} \sum_{n=1}^N \eta_n = \infty$. This means that as long as the step-sizes gradually shrink (allowing the algorithm to settle) but their sum remains infinite (ensuring it can still explore the entire space), the discrete algorithm will successfully approximate the true target distribution.

2.3 TEMPERATURE SCALING

Building upon the stochastic optimization framework, (Harel et al., 2025) analyze Continuous Langevin Dynamics (CLD) as a fundamental model for understanding generalization in over-parameterized neural networks. Unlike traditional analyses of Stochastic Gradient Langevin Dynamics (SGLD) that focus on the discrete-step updates introduced by Welling and Teh, this work prioritizes the continuous-time process described by the Stochastic Differential Equation (SDE):

$$dX_t = -\nabla L_S(X_t)dt - \lambda\beta^{-1}X_tdt + \sqrt{2\beta^{-1}}d\mathbf{w}_t \quad (6)$$

where the parameter evolution is governed by the training loss gradient, a regularization term, and Gaussian noise scaled by the inverse temperature β . The central theoretical breakthrough lies in the observation that for any Markov process with a Gibbs-type stationary distribution $p_\infty(X) \propto \exp(-\Psi)$, where the potential is typically $\Psi = \beta L_S$, the marginal distribution's divergence from the initialization remains bounded. This monotonicity, implied by a generalized second law of thermodynamics, allows for the derivation of a dimension-independent generalization bound that remains valid at any intermediate training step $t \geq 0$ without requiring the process to reach equilibrium. Formally, for a sample size N and with probability $1 - \delta$ over the dataset $S \sim D^N$, let $\Delta_t = \mathbb{E}_{X_t \sim p_t}[E_D(X_t) - E_S(X_t)|S]$ denote the expected generalization gap. This gap is bounded by the following inequality:

$$\Delta_t \leq \sqrt{\frac{\beta \mathbb{E}_{X \sim p_0}[L_S(X)|S] + \ln(N/\delta)}{2N}} \quad (7)$$

This formulation demonstrates that if the inverse temperature follows the scaling law $\beta = O(N)$, the thermal noise effectively controls model complexity, providing a guarantee that is completely independent of the model's dimensionality d or the specific norms of the gradients.

3 CONTRIBUTIONS & GOALS

The theoretical landscape of Stochastic Gradient Langevin Dynamics (SGLD) has largely been divided into two separate research tracks. On one hand, studies on convergence speed emphasize the benefits of non-Gaussian noise and fractional dynamics for escaping local minima. On the other hand, recent advances in generalization theory, notably the framework proposed by (Harel et al., 2025), show that temperature dynamics play a decisive role in controlling the generalization gap regardless of the model dimension. However, a significant gap remains: current generalization bounds are often distribution-agnostic. By assuming a worst-case Gaussian structure to maintain generality, these bounds may fail to capture the tighter limits achievable when the stochastic gradient noise exhibits heavy-tailed behavior.

In this project, we aim to bridge these two perspectives by proposing that the tail-index α , which characterizes the jumpiness of the noise, is not just a tool for optimization speed but also a critical factor in determining the required noise magnitude for generalization. Our main goals are as follows:

Fractional Temperature Scaling and Generalization Bounds: We aim to formalize tighter generalization bounds by incorporating the tail index α into existing temperature scaling laws. Gaussian based theories, such as those in (Harel et al., 2025), often yield conservative bounds because standard Brownian motion relies heavily on high temperatures to escape sharp local minima. In contrast, the Lévy jumps associated with fractional noise inherently accelerate these exit times. Leveraging this

property, our goal is to extend the temperature scaling framework to heavy-tailed dynamics and derive tighter bounds that explicitly account for the tail index α .

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A PROOF OF THEOREM 1

Proof. Let us define $q(X, t)$ as the probability density function of the state X at time t . We obtain the equation associated with the SDE as follows:

$$\partial_t q(X, t) = -\partial_X [b(X, \alpha) q(X, t)] - \mathcal{D}^\alpha q(X, t). \quad (8)$$

By using the definition of $b(X, \alpha)$, we obtain:

$$\begin{aligned} \partial_t q(X, t) &= -\partial_X \left[\frac{\mathcal{D}^{\alpha-2} \{-\phi(X) \partial_X U(X)\}}{\phi(X)} q(X, t) \right] - \mathcal{D}^\alpha q(X, t) \\ &= -\partial_X \left[\frac{\mathcal{D}^{\alpha-2} \{-\pi(X) \partial_X U(X)\}}{\pi(X)} q(X, t) \right] - \mathcal{D}^\alpha q(X, t). \end{aligned} \quad (9)$$

Here, we used the fact that $\pi(X) = \phi(X)/Z$, where $Z = \int \phi(X) dX$. By using $-\partial_X U(X) = \partial_X \log \pi(X) = \frac{\partial_X \pi(X)}{\pi(X)}$, we obtain:

$$\partial_t q(X, t) = -\partial_X \left[\frac{\mathcal{D}^{\alpha-2} \{\partial_X \pi(X)\}}{\pi(X)} q(X, t) \right] - \mathcal{D}^\alpha q(X, t). \quad (10)$$

We can verify that $\pi(X)$ is an invariant measure of the Markov process $(X_t)_{t \geq 0}$ by checking if the derivative is zero when $q(X, t) = \pi(X)$:

$$\begin{aligned} & -\partial_X \left[\frac{\mathcal{D}^{\alpha-2}\{\partial_X \pi(X)\}}{\pi(X)} \pi(X) \right] - \mathcal{D}^\alpha \pi(X) \\ &= -\partial_X [\mathcal{D}^{\alpha-2}\{\partial_X \pi(X)\}] - \mathcal{D}^\alpha \pi(X) \end{aligned} \tag{11}$$

$$= -\partial_X^2 [\mathcal{D}^{\alpha-2}\{\pi(X)\}] - \mathcal{D}^\alpha \pi(X) \tag{12}$$

$$= \mathcal{D}^2 [\mathcal{D}^{\alpha-2}\{\pi(X)\}] - \mathcal{D}^\alpha \pi(X) \tag{13}$$

$$= \mathcal{D}^\alpha \{\pi(X)\} - \mathcal{D}^\alpha \{\pi(X)\} \tag{14}$$

$$= 0. \tag{15}$$

If $b(X, \alpha)$ is Lipschitz continuous, we can conclude that $\pi(X)$ is the unique invariant measure of the Markov process $(X_t)_{t \geq 0}$. $\square$